AI Forward Deployed Engineer

The AI Architect & FDE Course



Week 1 Day 1

Welcome to the most important course I've made



An estimated
30,000
job openings globally

Salaries up to
$400,000
Potential upside over $1m for a senior top performer

This course is the roadmap

Before we start

First let's cover objectives, introduction, curriculum, logistics



Instant gratification

An agent harness from first principles



3 weeks to Forward Deployed Engineer

Theory Practical Projects

Week 3: Become an AI FDE



AI Solutions Architect	Security, Guardrails, API Gateways	Enterprise Context Layer & Integrations	Production Deployment	Capstone

Week 2: Become an AI Architect

Architecture, Evals, Observability	Agentic RAG, MCP, Memory	Durable State & Human in the Loop	Agent Teams, Swarms, Fleet	Self-Improvement, RSI

Week 1: Become an AI Engineer

Path to FDE	Frontier LLMs & Agents	Coding Agents & ADLC	Open-Source, Local, Token Cost	Harness Engineering

How this course fits in

Technical & Non-technical			Technical & Aspiring technical		Technical
1	**2**	**3**	**4**	**5**	**6**
AI Builder	**AI Coder**	**AI Leader**	**AI Engineer** Core Track	**AI Engineer** Agentic Track	**AI Engineer** Production Track
					
Agents Voice Agents n8n ElevenLabs	Claude Code Copilot Codex Antigravity	AI Delivery AI Transformation Start an AI business	LLMs Open-Source RAG LangChain Fine-tune	Agent loops OpenAI Agents SDK CrewAI LangGraph MCP	Vercel AWS, Bedrock GCP Azure LangFuse
Deliver agents for yourself and your clients	**Rapidly deliver Software with Coding Agents**	**Deliver commercial impact with AI**	**Select, Apply and Optimize LLMs**	**Build Autonomous Agents**	**Deploy LLMs and Agents at scale**

AI-Native Program

AI-Native Professional	**AI-Native Developer**	**AI Forward Deployed Engineer**
AI Fluency, LLMs and Agents	Coding Agents, Harnesses and the ADLC	The AI Architect & FDE Course

Who this course is for

Where are you starting from?



New to Python/LLMs

AI Engineer in training

Proficient AI Engineer

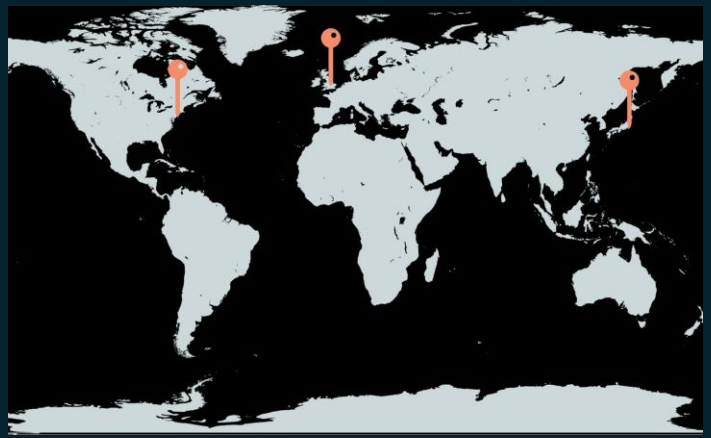
AI Architect

The sweet spot for this course

New to Python/LLMs
Also a good fit
Week 1 might be intense

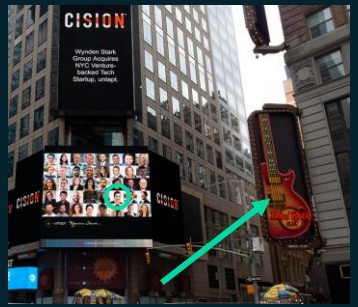
AI Architect
Also a good fit
You might want Week 1 on 2X

Introducing me











The origin story

The backstory to the FDE



Classical FDE

Palantir engineers on site with a customer to make the platform work

Shyam Sankar, Palantir CTO, credited with pioneering the term in 2007



The OpenAI definition

Applied AI engineers embedded with customers to build on OpenAI's models

May 2026: OpenAI launches company to embed “FDEs into organizations working on complex problems in demanding environments”



Broader industry usage

The term is now being used more broadly for Applied AI Engineering

In theory vs in practice

How FDE Engineer is used

In theory - NARROW (OpenAI)

Senior Applied AI Engineer embedded with a client

An FDE Engineer is an applied AI engineer embedded with a client and often working on site, responsible for getting AI solutions into production

In practice - BROADER

Applied AI Engineer solving commercial problems

An FDE Engineer is an applied AI engineer solving commercial problems by selecting, applying and deploying LLMs and Agents

What it takes to become an AI FDE

Three disciplines



AI Engineering



AI Architecture



Forward Deployed Engineering

Two areas of hands-on experience



Solved real-world business problems



Delivered to customers

The three disciplines

Week 1

AI Engineering

Agent loops and tools
Frontier LLMs and agents
Model selection
Agent frameworks
Coding agents
The ADLC
Open-source & local models
Token cost
Context engineering
Harness engineering

Technologies

OpenAI SDK
OpenRouter
LangChain
Claude Code
Claude Agent SDK

Week 2

AI Architecture

System design
Agent architecture
Evals and observability
Cost and speed trade-offs
Agentic RAG and memory
MCP
Durable state
Human in the loop
Agent swarms
Self-improvement

Technologies

Langfuse
Qdrant / vector db
MCP
LangGraph
Supabase

Week 3

Forward Deployed Engineering

Customer discovery
Governance & compliance
Security and guardrails
API and MCP gateways
Enterprise context layer
Enterprise integrations
Cloud infrastructure
Production deployment
Operational handover
Adoption

Technologies

Serverless
AWS AgentCore Gateway
AWS AgentCore Policy
AWS AgentCore Memory
AWS AgentCore Runtime

Two areas of hands-on experience

Adjacent to the FDE



Solved real-world business problems

What it means
You solved a commercial problem using LLMs or agents

Example
An internal RAG project that your company uses

On the course
You automate travel claims for a fictional insurer



Delivered to customers

What it means
You worked directly with a customer on a technical project

Example
Technical consulting or a project for an internal customer

On the course
You take the insurer's project from pilot to handover

The next step after the course is hands-on experience in a role

The five areas in job adverts

Share of 316 job adverts that mention each area



88% AI Engineering

79% AI Architecture

95% Forward Deployed Engineering

88% Solved real-world business problems

78% Delivered to customers

How many of the five areas each advert mentions



41% All five areas

48% Four areas

10% Two or three

Nine in ten adverts mention at least four of the five areas

Source: our review of 316 adverts from 184 employers for FDE, AI Architect and related roles

Installations

Four tools to install before we start



VS Code
Microsoft's free code editor, where we write and run our code



Git
Version control so we can clone the repo and track changes



Node
JavaScript runtime, used to install coding agents and MCP servers



uv
Python package manager for the project (it installs Python too)

Now let's go do this for Windows PC and Mac

VS Code extensions

Two extensions to add to VS Code



Python
Microsoft's extension for running and debugging Python



Jupyter
Microsoft's extension for running notebooks inside VS Code

Both install from the Extensions view in VS Code

Setting up our repo

Clone the course repo and open it in VS Code

1. **Open a Terminal in VS Code**
   Terminal menu, then New Terminal

2. **Clone the repo**

```
git clone https://github.com/ed-donner/undercover
```

3. **Open the project**
   File menu, Open Folder, then choose the undercover folder

Then we'll set up the environment

Setting up the environment

Install the packages, add your key and pick the kernel

1

Install the packages

Run `uv sync` in the terminal; it creates the project's Python environment

2

Add your OpenRouter key to a .env file

```
OPENROUTER_API_KEY=sk-or-v1-...
```

3

Open the notebook and pick the kernel

Select Kernel, then Python Environments, then the project's .venv

Then we'll make our first LLM call

To the lab

Our first LLM call from the course repo



API costs

What should you expect to spend?

$10

The typical experience

Around $10 of API spend, all in

$0

The free route

Free alternatives throughout;
you may need more patience with free models

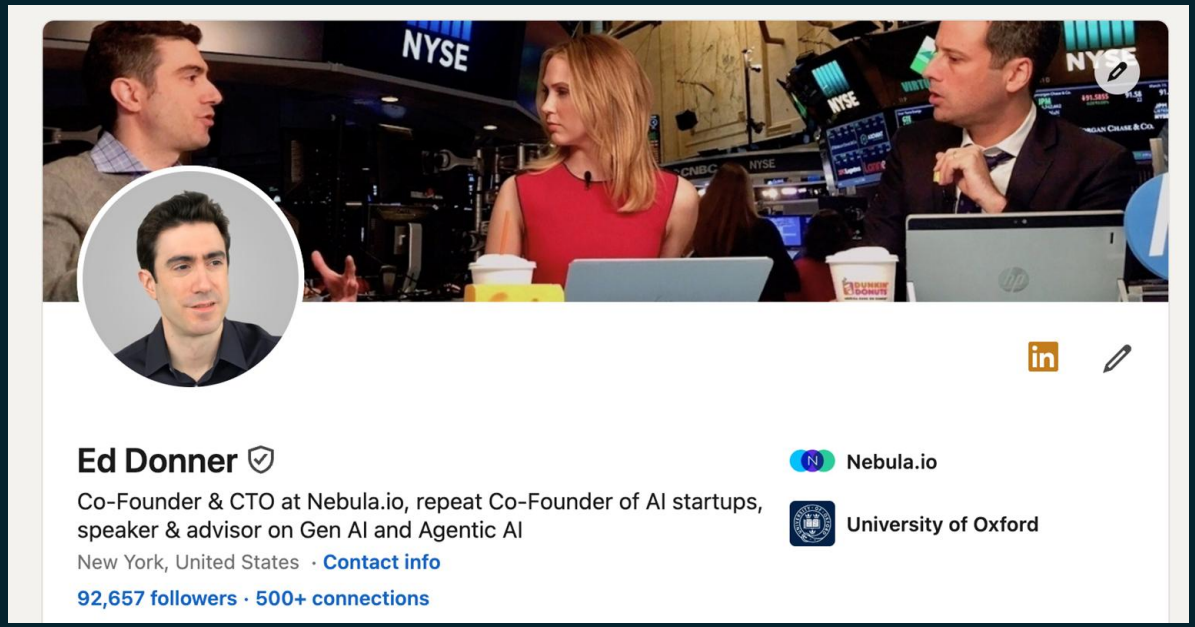
$20+

The full experience

Production infrastructure and more test data,
if you have the budget

I am here to help

https://www.linkedin.com/in/eddonner/



3 weeks to Forward Deployed Engineer

Theory Practical Projects

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7% Complete